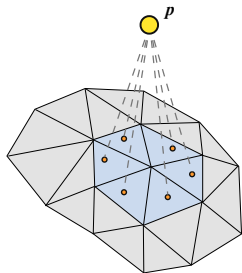
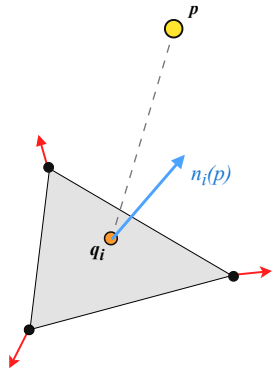


Global Geometric Guidance Field

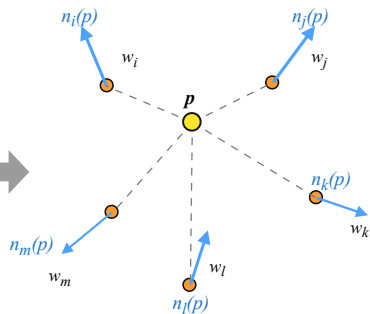
① K -nearest triangles



② projection + barycentric normal interpolation



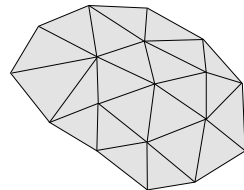
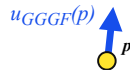
③ inverse-distance weighted fusion



$$u_{GGGF}(p) = \text{normalize}(\sum_i w_i n_i(p))$$

$$d(p) = \sum_i w_i d_i$$

④ final guidance direction



Inner CAD surface

p query point

q_i projection point

vertex normals

$n_i(p)$ local interpolated normal

$u_{GGGF}(p)$ final guidance vector